

Due: October 1, 2026

Remember to show your work for each problem to receive full credit.

Problem 0

Finish Problem 4 from HW 1.

See HW 1.

Problem 1

Suppose that we have an algorithm that takes as input a string of n bits. We are told that if the input bits are chosen independently and uniformly at random, the expected running time is $O(n^2)$. What can Markov's inequality tell us about the worst-case running time of this algorithm on inputs of size n ? [**Hint:** What is the sample space? What is the smallest probability of any event in that sample space?]

Let X be the random variable representing the running time of the algorithm on a random input of size n , so $\mathbb{E}[X] = O(n^2)$.

Let t be the worst-case running time of the algorithm on inputs of size n , so $\mathbb{P}(X \geq t) \geq \frac{1}{2^n}$ since the n bits are chosen independently and uniformly at random.

Markov's inequality states that for a non-negative random variable X and any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

so in particular,

$$\frac{1}{2^n} \leq \mathbb{P}(X \geq t) \leq \frac{O(n^2)}{t}$$

which means

$$t \leq O(n^2)2^n$$

Problem 2

- a. Define a random variable X and a value c for which the Markov inequality is tight, i.e. $\Pr(X \geq c\mathbb{E}[X]) = \frac{1}{c}$. [Hint: you can use a two point random variable, i.e. a random variable that gives non-zero probability to only 2 values.]

Let $X \sim \text{Bernoulli}(1/2)$. Then $\mathbb{E}[X] = 1/2$ and letting $c = 2$, we have

$$\mathbb{P}(X \geq 1) = \mathbb{P}(X > 1) + \mathbb{P}(X = 1) = \frac{1}{2}$$

- b. Define a random variable X and a value c for which the Chebyshev inequality is tight, i.e. $\Pr(|X - \mathbb{E}[X]| \geq c) = \frac{\text{Var}(X)}{c^2}$. [Hint: you can use a two point random variable with expectation 0.]

Let $Y \sim \text{Bernoulli}(1/2)$ and let $X = Y - \frac{1}{2}$. Then $\mathbb{E}[X] = \mathbb{E}[Y] - \mathbb{E}[1/2] = 0$ and $\text{Var}(Y) = \text{Var}(X) = 1/4$. Let $c = 1/2$. Then

$$\begin{aligned} \mathbb{P}(|X - \mathbb{E}[X]| \geq \frac{1}{2}) &= \mathbb{P}(X - \mathbb{E}[X] \geq \frac{1}{2}) + \mathbb{P}(X - \mathbb{E}[X] \leq -\frac{1}{2}) \\ &= \mathbb{P}(X \geq \frac{1}{2}) + \mathbb{P}(X \leq -\frac{1}{2}) \\ &= \mathbb{P}(X = \frac{1}{2}) + \mathbb{P}(X = -\frac{1}{2}) \\ &= \frac{1}{2} + \frac{1}{2} = 1 = \frac{1/4}{(1/2)^2} \\ &= \frac{\text{Var}(X)}{c^2} \end{aligned}$$

- c. Show an example for which the Markov inequality is not tight. [Hint: you can use a (continuous or discrete) uniform random variable.]

Let $X \sim \text{Uniform}(0, 1)$. Then $\mathbb{E}[X] = 1/2$. However, by Markov's inequality, we have

$$\mathbb{P}(X \geq \frac{1}{2}) = \frac{1}{2} \leq \frac{\mathbb{E}[X]}{1/2} = 1$$

- d. Show an example for which the Chebyshev inequality is not tight. [Hint: you can use a (continuous or discrete) uniform random variable with expectation 0.]

Let $X \sim \text{Uniform}(-1, 1)$. Then $\mathbb{E}[X] = 0$ and $\text{Var}(X) = 1/3$. However, by Chebyshev's inequality, we have

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq 1) = \mathbb{P}(|X| \geq 1) = 0 \leq \frac{\text{Var}(X)}{1^2} = \frac{1}{3}$$

Problem 3

Suppose that each draw independently produces one of n coupon types, chosen uniformly at random, where $n \geq 4$ is even. For a positive integer m , define $H_m = \sum_{i=1}^m \frac{1}{i}$. You may use the bound $\ln m \leq H_m \leq \ln m + 1$.

Let X be the number of draws until we have collected $n/2$ distinct coupon types. Before any draws are made, fix a particular set of $n/2$ coupon types, and let Y be the number of draws until we have collected every coupon type in this fixed set.

- a. Compute $\mathbb{E}[X]$ and $\mathbb{E}[Y]$ in terms of H_n and $H_{n/2}$. Give the asymptotic order of each expectation as $n \rightarrow \infty$.

First notice Y is the sum of $n/2$ geometric random variables representing the number of draws needed to collect each of the $n/2$ coupon types in the fixed set, so the i -th geometric random variable has success probability $\frac{n/2-i+1}{n}$. Therefore, we have

$$\mathbb{E}[Y] = \sum_{i=1}^{n/2} \frac{n}{n/2 - i + 1} = n \sum_{j=1}^{n/2} \frac{1}{j} = nH_{n/2} = \Theta(n \log n)$$

where $j = n/2 - i + 1$ and j goes from $n/2$ to 1 as i goes from 1 to $n/2$.

Similarly, X is the sum of $n/2$ geometric random variables representing the number of draws needed to collect each of the $n/2$ distinct coupon types but now the i -th geometric random variable has success probability $\frac{n-i+1}{n}$ since it suffices to collect any of the remaining $n - i + 1$ coupon types. Therefore, we have

$$\mathbb{E}[X] = \sum_{i=1}^{n/2} \frac{n}{n - i + 1} = n \sum_{j=n/2}^n \frac{1}{j} = n(H_n - H_{n/2}) = \Theta(n \log 2) = \Theta(n)$$

where $j = n - i + 1$ and j goes from n to $n/2$ as i goes from 1 to $n/2$.

- b. Express Y as a sum of independent geometric random variables. Compute $\text{Var}(Y)$ and show that

$$\text{Var}(Y) \leq \frac{\pi^2 n^2}{6}.$$

You may use $\sum_{i=1}^{\infty} \frac{1}{i^2} = \frac{\pi^2}{6}$.

We showed above that $Y = \sum_{i=1}^{n/2} Y_i$ where Y_i is a geometric random variable with success

probability $\frac{n/2-i+1}{n}$. Therefore, we have

$$\begin{aligned}
 \text{Var}(Y) &= \sum_{i=1}^{n/2} \text{Var}(Y_i) \\
 &= \sum_{i=1}^{n/2} \frac{1 - \frac{n/2-i+1}{n}}{\left(\frac{n/2-i+1}{n}\right)^2} \\
 &= \sum_{i=1}^{n/2} \frac{n - (n/2 - i + 1)}{(n/2 - i + 1)^2} n \\
 &= \sum_{i=1}^{n/2} \frac{n/2 + i - 1}{(n/2 - i + 1)^2} n \\
 &\leq \sum_{i=1}^{n/2} \frac{n/2 + n/2 - 1}{(n/2 - i + 1)^2} n \\
 &\leq \sum_{i=1}^{n/2} \frac{n}{(n/2 - i + 1)^2} n \\
 &= n^2 \sum_{j=1}^{n/2} \frac{1}{j^2} \leq n^2 \sum_{j=1}^{\infty} \frac{1}{j^2} = n^2 \frac{\pi^2}{6} = \frac{\pi^2 n^2}{6}
 \end{aligned}$$

where $j = n/2 - i + 1$.

- c. Let $t_n = \lceil 2nH_{n/2} \rceil$. Use Markov's inequality and Chebyshev's inequality separately to give upper bounds on $\Pr(Y > t_n)$. Compare the asymptotic orders of the two bounds.

From Markov's inequality, we have

$$\mathbb{P}(Y > t_n) \leq \frac{\mathbb{E}[Y]}{t_n} \leq \frac{nH_{n/2}}{2nH_{n/2}} = \frac{1}{2} = O(1)$$

From Chebyshev's inequality, we have

$$\begin{aligned}
 \mathbb{P}(Y > t_n) &= \mathbb{P}(Y - \mathbb{E}[Y] > t_n - \mathbb{E}[Y]) \\
 &\leq \mathbb{P}(|Y - \mathbb{E}[Y]| > t_n - \mathbb{E}[Y]) \\
 &\leq \frac{\text{Var}(Y)}{(t_n - \mathbb{E}[Y])^2} \\
 &\leq \frac{\pi^2 n^2 / 6}{(2nH_{n/2} - nH_{n/2})^2} = \frac{\pi^2 n^2 / 6}{(nH_{n/2})^2} = \frac{\pi^2}{6H_{n/2}^2} = O\left(\frac{1}{\log^2 n}\right)
 \end{aligned}$$

- d. Give a direct bound on $\Pr(Y > t)$ by considering, for each coupon type in the fixed set, the event that it has not appeared in the first t draws. Prove that for every integer $t \geq 0$,

$$\Pr(Y > t) \leq \frac{n}{2} \left(1 - \frac{1}{n}\right)^t \leq \frac{n}{2} e^{-t/n}.$$

Apply this bound with $t = t_n$ and compare it with the bounds obtained in part (c).

First, let A_i be the event that the i -th coupon type in the fixed set did not appear in the first t draws. Then we have

$$\begin{aligned} P(Y > t) &= \mathbb{P}\left(\bigcup_{i=1}^{n/2} A_i\right) \\ &\leq \sum_{i=1}^{n/2} \mathbb{P}(A_i) \\ &\leq \sum_{i=1}^{n/2} \left(1 - \frac{1}{n}\right)^t \\ &\leq \frac{n}{2} \left(1 - \frac{1}{n}\right)^t \\ &\leq \frac{n}{2} e^{-t/n} \end{aligned}$$

where the final line follows from the fact that $1 - x \leq e^{-x}$ for all x .

Then with $t = t_n$,

$$P(Y > t_n) \leq \frac{n}{2} e^{-t_n/n} \leq \frac{n}{2} e^{-2H_{n/2}} \leq \frac{n}{2} e^{-2(\log n/2)} \leq \frac{n}{2} \cdot \frac{4}{n^2} = \frac{2}{n} = O\left(\frac{1}{n}\right)$$

So the bound from part (d) is asymptotically tighter than the bounds from part (c).

Problem 4

Company G hires new employees only if they are better than their top current employees. The company scores employees in a range $[0, 1]$, where 0 is the best possible employee and 1 is the worst. The scores of candidates interviewed by the company are independent and uniformly distributed in the range $[0, 1]$.

Let r_i denote the score of the i -th hired employee in the company, and assume that r_1 was distributed uniformly in the range $[0, 1]$.

1. Compute $E[r_2]$.

We know $r_2 \sim \text{Unif}(0, r_1)$ and $r_1 \sim \text{Unif}(0, 1)$. Therefore, we have

$$\mathbb{E}[r_2] = \mathbb{E}_{r_1}[\mathbb{E}_{r_2}[r_2|r_1]] = \mathbb{E}\left[\frac{r_1}{2}\right] = \frac{1}{2}\mathbb{E}[r_1] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

2. Compute $E[r_k]$

Given $r_1, \dots, r_{k-1}$, we know that $r_k \sim \text{Unif}(0, r_{k-1})$. So,

$$\mathbb{E}[r_k] = \mathbb{E}_{r_{k-1}}[\mathbb{E}_{r_k}[r_k|r_{k-1}]] = \mathbb{E}\left[\frac{r_{k-1}}{2}\right] = \frac{1}{2}\mathbb{E}[r_{k-1}] = \frac{1}{2^k}$$

3. Let T_{k+1} be the number of candidates interviewed for the $k+1$ hire. It's not easy to compute $E[T_{k+1}] = E[\frac{1}{r_k}]$. Use Markov inequality (on r_k) to prove a lower bound 2^{k-2} for this expectation.

First notice

$$\mathbb{P}(r_k \geq 2^{-(k-1)}) \leq \frac{\mathbb{E}[r_k]}{2^{-(k-1)}} = \frac{2^{-k}}{2^{-(k-1)}} = \frac{1}{2}$$

so

$$\mathbb{E}[1/r_k] \geq 2^{k-1} \cdot \mathbb{P}\left(\frac{1}{r_k} \geq 2^{k-1}\right) = \frac{1}{2} \cdot 2^{k-1} = 2^{k-2}$$